INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

PROPELLANT GRAIN, MK43, MOD 1

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of propellent grain designated MK43 Mod 1. The grain consists of an extruded hollow cylinder made from N-5 sheet propellant with ethyl cellulose inhibiting material cemented on its periphery and both ends. It also has an end sleeve cemented on one end to facilitate a gas-tight seal between the grain and the motor nozzle.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition, General Specification for
MIL-A-3167	-	Adhesives (For Plastic Inhibitors)
MIL-P-17689	-	Propellant, Double-Base Sheet, Type N-5
MIL-F-63463	-	Film, Inhibiting, Ethyl Cellulose

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168	-	Ammunition Lot Numbering
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

9239050	-	Propellant Grain, MK43, Mod 1
9225665	-	Special, Ring Gage
1350840	-	Pallet Crate MK2, Mod 0
1350857	-	Propellant Grain Container (Inner) MK10, Mod 1

PUBLICATIONS

NAVY

OD9153	-	Internal Inspection of Rocket Propellant Grains

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(Copies of the drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Material. Materials and parts shall be in accordance with applicable drawings and specifications. Also, materials and parts shall be free from defects, including all surface imperfections and discontinuities and all internal defects in the finished component, resulting from flaws or lack of soundness in the materials used.

3.3 Propellant grain blank. The finished propellant shall be ready for attachment of the end inhibitors. Specifically, it shall be full formed, have smooth solid ends, be free of fissures and surface holes, be free of cracks and foreign materials, and show good workmanship throughout its entire length. It shall also meet all dimensional and physical requirements. The propellant contained in the grain blank shall conform to MIL-P-17689. Propellant manufactured and extruded on the same premises shall not be submitted and accepted in lots before extrusion. The Taliani Stability Test shall be performed by, or for, the Government on representative samples of propellant before extrusion or on the extruded powder at any stage of processing. The propellant in the end item shall have met the requirements of the stability test before the end item is presented for acceptance under this specification. All other requirements of the propellant specification shall still apply. Furthermore, information on the other test requirements of 3.3 of MIL-P-17689 shall be furnished by the contractor for each quantity of propellant on which the Taliani Stability Test was made.

3.3.1 Alternate propellant grain blank. Manufacturers of MK43 propellant grains may present to the procuring activity an alternate acceptance plan for N-5 propellant. The plan must state how and what parameters will be controlled to produce MK43 propellant grains meeting MIL-P-18811 requirements. The local contracting officer's representative may approve the quality assurance plan when directed by the procuring activity.

3.4 Internal defects. The propellant grain blank and assembled grain shall be free from internal foreign material and internal fissures.

3.5 Static firing. The grains of the sample selected for static firing performance testing shall be statically fired within 120 days of peripheral

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inhibiting. An adequate minimum drying time after inhibiting grains must be allowed to assure a proper bond before testing. The test stand and recording equipment shall be approved by the procuring activity.

3.6 Pressure time curves. Pressure time curves, from the application of the firing pulse until the gage pressure finally falls below 250 pound force per square inch (lbf/in^2), shall lie entirely between the inner and outer rectangles formed by the base line and the appropriate dotted lines of figure 1.

3.7 Inhibiting film. The inhibiting film shall conform to the requirements of MIL-F-63463.

3.8 Adhesive (ethyl lactate-butyl acetate). This adhesive shall consist of a homogeneous mixture of 31 to 35 percent ethyl lactate, by volume, and the balance of normal butyl acetate. The above proportions of these solvents are not as listed in MIL-A-3167. Excess solvent (runoff solvent) collected from the peripheral inhibiting of propellant grains of N-5 composition with ethyl cellulose may be reused, provided that fresh solvent is added as necessary to limit the non-volatile matter to be not greater than 0.9 grams/100 milliliters (ml).

3.9 Room temperature. The room temperature shall be held at 21 $\pm$ 3°C during the time dimensional acceptance measurements are made. Samples shall be conditioned at the above temperature for 4 hours prior to measuring, or for a period of time to assure to the contracting officer that the temperature equilibrium has been reached.

3.10 Annealing. The grain blank shall be annealed for not less than 8 hours and held at not less than 60°C. The annealing treatment process shall be approved by the procuring activity.

3.11 Air bath test. The end sleeve shall be capable of withstanding, without cracking, the following air bath temperature cycles repeated two times: not less than 6 hours at -54 $\pm$ 3°C, followed immediately by not less than 6 hours at 73 $\pm$ 3°C.

3.12 Workmanship. The propellant grain shall conform to the best commercial practices to assure a quality production item that complies with applicable drawing dimensions, finishes and tolerances. It shall be free of dirt, foreign material or other contaminants and shall be formulated to meet the requirements of this specification. All markings shall be neatly and sharply defined.

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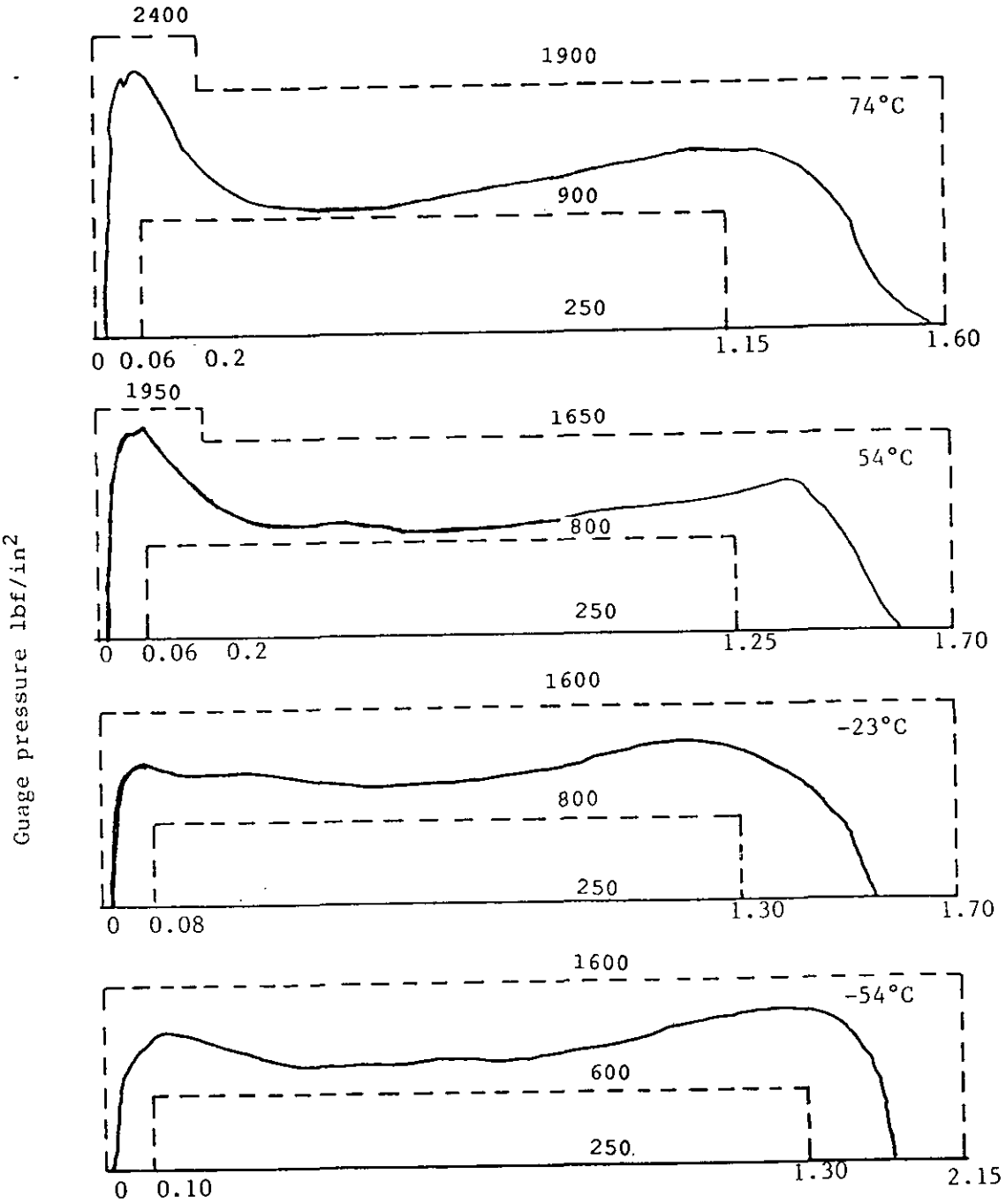


FIGURE 1. Gauge pressure time curve limits Bourdon or electronic recorder propellant grain MK43 mods 0 and 1.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with 4.6.3.

4.4 First article inspection.

4.4.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.4.2. The first article sample shall consist of 140 assemblies and 140 complete sets of parts (for example, each 140 of every component and/or subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using this same production process, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a

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record of this inspection with the sample, including statements of findings and/or certificates of conformance for materials (see 4.6.1c). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period greater than 90 days, or whenever change occurs in manufacturing process, material used, drawings or specification such as to significantly affect product uniformity as determined by the Government.

4.4.2 Inspections to be performed. Assemblies and components shall be subjected by the Government to any or all of the examinations or tests specified in 4.6.3 and 4.8 of this specification and any or all requirements of the applicable drawings.

4.4.3 Rejection. If any assembly or component fails to comply with any of the requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly or component in the sample to comply with any of the stated requirements.

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table I, the applicable drawings as specified herein, and as specified by the contracting officer (see 6.2).

TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Material	3.2	4.6.1
Internal defects	3.4	4.8.3
Static firing	3.5	4.8.4
Pressure time curves	3.6	4.8.4.1.3
Adhesive (ethyl lactate-butyl acetate)	3.8	4.8.2
Air bath test	3.11	4.8.1
Workmanship	3.12	4.8.1, 4.9.1

4.6 Inspection provisions for production.

4.6.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply as specified in the contract or order (see 6.2), the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

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b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review. Certificates of conformance shall be submitted by the contractor as follows:

(1) A statement by the contractor that each component of all N-5 propellant used in the manufacture of a MK43 grain lot, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, conforms to the provisions of those specifications. This certification shall be of a general nature, simply indicating that all such materials conform to the applicable specifications. Copies of original certificates of conformance or acceptance data submitted to the MK43 producers by the suppliers of the material shall not be submitted to the Government. The MK43 producer shall, however, have this information available for presentation to the Government upon request.

(2) A certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority shall not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization. A contractor shall not knowingly include defective units in a lot. The contractor shall furnish the above certificates of conformance along with all data required for N-5 propellant by MIL-P-17689. As stated in paragraph 3.3., all test data for N-5 propellant used in each MK43 lot shall be furnished to the Government by the contractor.

d. Number of items in the lot.

e. Date submitted.

4.6.2 Lot formation. In this specification the term inspection lot is an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the inspection

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lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be 20,000 ± 500 or 40,000 ± 500. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor and the Government inspector.

4.6.3 Examination. The examinations specified herein shall be performed on inspection lots (see 4.6.2) of the components or subassemblies in the applicable paragraph headings. Inspection for critical defects (and major defects when so specified) shall be 100 percent.

a. Sampling Plans. Sampling plans shall be in accordance with Tables II, III, IV, V and VI of this specification. Sampling Plans for major defects shall be in accordance with sampling plans specified herein, except that variable sampling plans in accordance with MIL-STD-414 may be used if requested by the contractor and approved by the contracting officer.

b. Critical Defects. Unless otherwise specified, inspection for critical defects shall be 100 percent.

c. Major Defects. When required, major defects shall be inspected 100 percent. Except where 100 percent inspection is specified, inspection for major defects shall be to the individual sampling plan tables indicated in the classification of defects and test paragraphs provided in this detail specification. The sampling plans for testing (see 4.8) shall be as specified.

d. Unlisted Defects. See MIL-A-2550.

e. Individual defects. Instead of applying the sampling plans designated herein to the individual defects, the group sampling plans listed for each major subparagraph in 4.6.3 may be used at the option of the contractor for those groups where a group sampling plan is assigned. Each group sampling plan applies to a major group indicated in the subparagraphs of paragraph 4.6.3, and if this method is elected, it shall be used for all of the subparagraphs of paragraph 4.6.3, where group sampling plans are so designated for this purpose.

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TABLE II. Sampling plan I.

Lot Size	Sample Size
2 to 13	All
14 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE III. Sampling plan II.

Lot Size	Sample Size
2 to 20	All
21 to 280	20
281 to 1200	47
1201 to 3200	53
3201 to 10,000	68
10,001 to 35,000	77
35,001 to 150,000	96
150,001 to 500,000	119
500,001 and over	143
In all cases: Acceptance number is ZERO Rejection number is ONE	

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TABLE IV. Sampling plan III.

Lot Size	Sample Size
2 to 500	All
501 to 150,000	500
150,001 to 500,000	750
500,001 and over	1112
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE V. Sampling plan IV.

Lot Size	Sample Size
2 to 15	2
16 to 25	3
26 to 90	5
91 to 150	6
151 to 280	7
281 to 500	9
501 to 1200	11
1201 to 3200	13
3201 and over	15
In all cases: Acceptance number is ZERO Rejection number is ONE	

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TABLE VI. Sampling plan V.

Lot Size	Sample Size
2 to 5	All
6 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3201 to 10,000	29
10,001 to 35,000	35
35,001 and over	40
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.7 Classification of defects. Each sample shall be tested for defects as specified in 4.7.1 thru 4.7.8.

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4.7.1 End sleeve, (see drawing 9240823, covering a detail of drawing 9239050).

TABLE VII. Classification of defects, End sleeve.

Categories	Defects	Method of Inspection	Sampling Plan (Tables)
Critical:	None defined.		
Major:			
101	Concentricity of inner tapered surface measured within 0.10 inch of plane S-S to small OD	Dial fixture gage	II
102	Concentricity of large outside diameter to small OD	Dial fixture gage	II
103	Flatness of surface X-X improper	Surface plate and 0.005 inch feeler stock	II
104	Small outside diameter	Plain ring gage "Go" and "No Go"	II
105	Evidence of imperfections on surface Y-Y and small OD up to 0.25 inch aft of surface Y-Y	Visual	II
106	Flatness of surface Y-Y improper	0.005" Feeler stock	II
107	Presence of bubbles	Visual	II
108	Evidence of imperfections on surface X-X	Visual	II
109	Evidence of weld lines	Visual	II
110	Air bath test	Air bath	II
111	Evidence of cracks or laminations	Visual	III

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4.7.2 Grain blank extruded, annealed and cut to length (see drawing 9240828, covering a detail of drawing 9239050).

TABLE VIII. Classification of defects, grain blank extruded, annealed and cut to length.

Categories	Defects	Method of Inspection	Sampling Plan (Tables)
Critical:	None defined		
Major:			
101	Major inside diameter for the first 0.5" inch from the end of the propellant grain, minimum	Special plug gage "GO"	II
102	Grain straightness incorrect	Surface plate feeler stock (0.120 inch)	II
103	Outside diameter, minimum	Adjustable Snap gage	II
104	Grain length incorrect	Length gage	II
105	Range of grain length per lot	Length gage	II
106	Perpendicularity, circumferential surface of grain blank to end surface of grain blank	Squareness gage	II

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4.7.3 End inhibitor (see drawing 9240822, covering a detail of drawing 9239050).

TABLE IX. Classification of defects, end inhibitor.

Categories	Defects	Method of Inspection	Sampling Plan (Tables)
Critical:	None defined		
Major:			
101	Surface flatness incorrect (both sides)	Surface plate feeler stock (0.005 inch)	II
102	Evidence of bubbles	Magnifier	II
103	Evidence of cracks or laminations	Visual	III

4.7.4 Machined grain (see drawing 9240824, covering a detail of drawing 9239050).

TABLE X. Classification of defects, machined grain.

Categories	Defects	Method of Inspection	Sampling Plan (Tables)
Critical:	None defined.		
Major:			
101	Grain straightness incorrect	Surface plate feeler stock (0.100 inch)	II
102	Difference between thickness and thinnest web thickness, maximum	Dial thickness gage	II
103	Outside diameter, maximum, functional	Special ring gage	II
104	Outside diameter, minimum	Build-up snap	II
105	Evidence of improper bonding of end inhibitors	Visual	II

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4.7.5 Inhibited grain (see drawing 9240829, covering a detail of drawing 9239050).

TABLE XI. Classification of defects, inhibited grain.

Categories	Defects	Method of Inspection	Sampling Plan (Tables)
Critical:	None defined.		
Major:			
101	Grain straightness improper	Straight edge, plug flat	II
102	Major inside diameter of grain perforation throughout the first 0.50 inch from plane Y-Y minimum	Special plug	II
103	Difference between thickest and thinnest overall web thickness measured from peripheral inhibitor, maximum	Dual thickness gage	II
104	Evidence of improper wrap	Visual	II
105	Evidence of improper bonding	Visual	II
106	Evidence of foreign material between inhibitors and propellant and between layers of inhibiting tape	Visual	II
107	Bonding between end inhibitor and propellant inadequate	Measuring magnifier	II
108	Radial thickness of peripheral inhibitor	Measuring magnifier	II

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4.7.6 Propellant grain, MK43 MOD 1 (see drawing 9239050).

TABLE XII. Classification of defects, propellant grain MK43 MOD 1.

Categories	Defects	Method of Inspection	Sampling Plan (Tables)
Critical:			
1	Outside diameter of propellant grain, minimum	See note 8 on drawing number 9239050	All
2	Evidence of cracks or laminations in peripheral inhibitor, end sleeve	Visual	All
Major:			
101	End sleeve concentricity and end sleeve length incorrect	Special ring gage, drawing 9225665 Rev A	II
102	Outside diameter of propellant grain, maximum, functional	Functional ring gage	II
103	Evidence of defects on outside surface of end sleeve between planes Y-Y and U-U	Measuring magnifier	II
104	Evidence of defects on top of Y-Y surface of end sleeve	Measuring magnifier	II
105	Evidence of defects on peripheral edge of Y-Y surface of end sleeve	Measuring magnifier	II
106	Bond between end sleeve and inhibited grain inadequate	Measuring magnifier	II
107	Excessive unbonded area between peripheral inhibitor and outer rim of end inhibitor	Measuring magnifier	II
108	Excessive unbonded area between propellant and peripheral inhibitor and between layers of inhibitor	Measuring magnifier	II
109	Static firing performance test		See 4.8.4

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4.7.7 Adhesive, (see drawing 9239050).

TABLE XIII. Classification of defects, adhesive.

Categories	Defects	Method of Inspection	Sampling Plan (Tables)
Critical:	None defined.		
Major: 101	Adhesive conformance test		See 4.8.2

4.7.8 Propellant grain MK43 MOD 1 or grain blank extruded annealed and cut to length (see drawing 9239050).

TABLE XIV. Classification of defects, propellant grain MK43 MOD 1.

Categories	Defects	Method of Inspection	Sampling Plan (Tables)
Critical:	None defined.		
Major: 101	Internal inspection test		See 4.8.3

4.8 Testing.

4.8.1 Air bath test Major Defect No. 110 (see table VII). Fifty end sleeves per lot shall be randomly selected and subjected to an air bath test as specified in 4.9.1 using equipment in accordance with 4.8.5. Any end sleeve exhibiting cracks shall be considered defective. One defective item shall be cause for lot rejection. This is considered a destructive test. Any item so tested shall not be returned to the lot.

4.8.2 Adhesive (ethyl-lactate, butyl-acetate) conformance test (see 3.8) Major Defect No. 101 (see table XIII). The ethyl lactate used shall conform to MIL-A-3167. The adhesive mixture shall conform to 3.8. This test shall be performed in accordance with 4.9.4 using equipment in accordance with 4.8.5.

4.8.3 Internal inspection test (see 3.4) Major Defect No. 101 (see table XIV). The propellant grain, defined as the item in any stage of production after extrusion, shall be subjected to fluoroscopic or X-ray inspection for the presence of internal fissures and foreign matter as specified in 3.4 and 4.9.3 using equipment in accordance with 4.8.5. Table I sampling plan shall apply to defectiveness as specified in 4.9.3. However, one longitudinal fissure which exceeds two inches in length or 0.125 inch in

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width, or one transverse fissure which exceeds 0.125 inch in length shall be cause for lot rejection.

4.8.4 Static firing performance test (see 3.5) Major Defect No. 109 (see table XII). This test shall be performed as specified in 3.5 and 4.9.2 using equipment in accordance with 4.8.5. If 1 motor failure occurs (which includes a high pressure blowup, or a burn through or fracture) or 3 or more incipient failures (as evidenced by bulges in the sides of the tubes at least 0.0625 inches high), the lot shall be rejected. If, as a result of reduced inspection, the total sample for static tests is reduced to 100 or less grains, the lot shall be rejected if 2 or more incipient failures occur, or 1 motor failure occurs, which is determined by the government to be attributable to the grain.

4.8.4.1 Sampling. The special sampling plan for this test shall be in accordance with MIL-STD-414. The sample sizes for such tests are the following:

Lot Size	Normal or Tightened	Reduced
20,000 ± 500	140X	60
40,000 ± 500	200XX	80

X - Code letter K, Table B-3, MIL-STD-414

XX - Code letter M, Table B-3, MIL-STD-414

The samples shall be divided evenly among the four preconditioning temperatures, 74°C, 54°C, -23°C, and -54°C.

Minimum sample sizes per test temperature (due to loss of test results - see 6.6.6).

Sample Size	Minimum Sample Size
50	48
35	33
20	19
15	14

4.8.4.1.2 Criteria for severity of inspection. The normal inspection initially in effect in accordance with MIL-STD-414, shall be changed to reduced inspection when the 4 parameters (early and late maximum pressure at 74°C, formula time at 54°C, and formula time at -23°C) are such that acceptable process quality indices may be calculated for a group of 10 consecutive lots. Instructions for calculating process indices are contained in MIL-STD-414. A

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change is required from reduced to normal inspection at a given temperature when the parameter associated with the temperature fails to meet the stated requirement. When this condition exists normal inspection will be reinstated at -54°C and at the temperature at which the parameter is not met for 10 consecutive lots.

4.8.4.1.3 Shape and smoothness of pressure-time curves. Each curve shall be examined by the Government inspector and classified as normal or defective based on 3.6 and figure 1. The conditions that would cause a pressure-time curve to be classified as defective are as follows:

a. Failure of the curve to meet the shape requirements of 3.6 and figure 1, except as specified in 4.8.4.1.4. For a lot to meet the requirements as to shape smoothness of its pressure-time curves, the total number of curves classified as defective among all its sample grains fired statically shall be not greater than 5. A defective curve shall be one which shows the presence of 1 or more defects as to shape or smoothness. Results of this inspection shall be recorded on the summary sheet. The total number of pressure-time curves classified as defective shall be not greater than 6, or the lot shall be rejected. The total number of curves classified as defective for reduced inspection shall be not greater than 3, or the lot shall be rejected; or

b. One or more sudden, abnormal, pressure rises equal to, or greater than 250 lbf/in2 for -23°C and 54°C firings; or

c. Nozzle stoppages or end pips (characterized by sudden pressure rise after the final maximum pressure) resulting in pressures exceeding 1900 lbf/in^2. A sudden rise in the pressure-time curve indicates a momentary nozzle stoppage. It indicates an abnormal condition within the motor, and calls for a thorough investigation of the cause of nozzle stoppage. Such stoppages near the end of the burning period may occur with these grains when fired at high temperatures after a prolonged storage, especially if part or all of the storage has been above 38°C.

4.8.4.1.4 Erratic pressure at 74°C. Erratic pressure pips (see 6.6.7) at 74°C only, shall be subject to a separate control based on severity. For an acceptable lot, no peak pressure shall exceed 225 lbf/in^2 and not more than 5 peak pressures shall exceed 1900 lbf/in^2 (such curves shall be counted as defective in accordance with 4.8.4.1.3a above). If the first ballistic sample fails to meet the requirement of this paragraph, a second sample of the same size shall be randomly selected for retest (at 74°C only). Retest shall take place only after not less than 21 calendar days from the date of the original test. Accept the lot if no peak pressure exceeds 225 lbf/in^2 and not more than 5 peak pressures exceed 1900 lbf/in^2, for the retest sample. If the retest sample fails to meet these requirements, the lot shall be rejected.

4.8.4.1.5 Early maximum pressure at 74°C. For an acceptable lot with regard to early maximum pressure at 74°C, the upper specification limit shall

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be 2400 lbf/in^2. Sampling plan I, table II shall be applied to this test requirement and evaluated in accordance with MIL-STD-414.

4.8.4.1.6 Late maximum pressure at 70°C. For an acceptable lot with regard to late maximum pressure at 74°C, the upper specification limit shall be 1900 lbf/in^2. Sampling plan I, table II shall be applied to this test requirement and evaluated in accordance with MIL-STD-414.

4.8.4.1.7 Formula time at -23°C and 54°C. For an acceptable lot with regard to formula time at -23°C and 54°C, the upper specification limit shall be 1.924 seconds and the lower specification limit shall be 1.804 seconds. Sampling plan I, table II shall be applied to this test requirement and evaluated in accordance with MIL-STD-414.

4.8.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.6.3 and 4.9.

4.9 Test methods and procedures.

4.9.1 Air bath test (see 4.8.1). The end sleeve shall be placed in an air-bath and shall be capable of withstanding, without cracking, the following air bath temperature cycle repeated 2 times: not less than 6 hours at -54°C ± 3°C followed immediately by not less than 6 hours at 74°C ± 3°. There shall be no delay between cycles.

4.9.2 Static firing performance test (see 4.8.4). Grains in the test motors shall be conditioned for not less than 8 hours and not greater than 30 hours. The temperature of the propellant grains shall be within ± 3°C of the nominal value during the last 3 hours of conditioning. If a recycle becomes necessary, return grain temperature to 21 ± 3°C for not less than 12 hours, and then repeat the normal temperature conditioning procedures. Recycling shall be noted in the acceptance report. Each firing shall be initiated within 2 minutes of the time the motor is removed from the temperature conditioned cabinet.

4.9.3 Internal inspection test (see 4.8.3). All fluoroscopic or radiographic examinations shall be performed and evaluated in accordance with OD9153. A standard defective grain shall be fluoroscoped not less than once per shift.

4.9.4 Adhesive (ethyl lactate-butyl acetate) conformance test. Test procedures to be used in evaluating the requirements of 4.8.2 shall be those specified in 4.3.1 to 4.3.9, inclusive, of MIL-A-3167.

4.10 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-STD-1169.

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5. PACKAGING

5.1 Packaging and packing. Propellant grains MK43 MOD1 shall be packaged in propellant grain container (inner), drawing 1350857, MK10 MOD1 and packed in pallet crate MK2 MOD 0, drawing 1350840. The propellant grain container MK10 MOD 0 may be used as an alternate container. Only grains from a single production lot shall be packed in any one box. Individual grain containers do not need to be marked in accordance with drawing 1350857. Each pallet crate shall have necessary information on a placard affixed to the side of the crate.

5.2 Marking. Each shipping container shall be marked in accordance with MIL-STD-129, in compliance with the marking requirements for Level C containers, except that the DOT marking shall be permanently affixed to the shipping container.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The propellant grain covered by this specification is intended for use in rocket motors as a sheet propellant with ethyl cellulose inhibiting material.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Drawings when required, providing dimensions, tolerances, hardness, and other design details (see 3.2)

d. When first article is required (see 3.1 and 4.4)

e. First article sample size (see 4.4.1)

f. Conditions for quality conformance inspection (see 4.5)

g. Requirements for preservation, packing and marking (see 4.10 and section 5)

h. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167, except when propellant or explosive lots are involved. Acceptance and Description Sheets shall be prepared in accordance with MIL-STD-1171.

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Submission of first article sample. Instructions as to the location for the evaluation of first article sample shall be obtained from the contracting officer. Delivery of test data, hardware and inspection records shall be in accordance with appropriate contract documents.

6.5 Data reporting. The following information will be reported in accordance with the provisions of the contract.

a. Process averages and standard deviations for the following variables defects will be reported:

(1) Early maximum pressure test at 74°C

(2) Late maximum pressure test at 74°C

(3) Formula-time test at -23°C

(4) Formula-time test at 54°C

b. The following attributes/defects shall be reported:

(1) Erratic pressure test at 74°C

(2) Taliani Stability Test on propellant

(3) Grain-ignition failures or misfires

(4) Motor failures

(5) Incipient failures

(6) Shape and smoothness of curves

(7) Attribute inspection

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(a) Critical

(b) Major

(c) Internal Inspection

c. All results of tests and inspections required by MIL-P-17689 shall also be reported.

d. Reporting format. The above information shall be submitted on a format deemed adequate by the Government inspector. The Government inspector is also requested to submit any other information he deems necessary for evaluation of the quality of the material.

6.6 Definitions.

6.6.1 Contractor. Unless otherwise specified, the word "Contractor", used throughout this specification, shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contracts, Government-Owned, Contractor-Operated (GOCO) and Government-Owned, Government-Operated (GOGO) plants.

6.6.2 Formula time (F.T.) Formula time (F.T.) is defined as a quantity, expressed in seconds to the nearest millisecond, which approximates the time required for the rocket to travel a certain distance under specified conditions. Formula times for individual grains are not required. The mean formula time is defined for either the -23°C or 54°C firings separately as:

$$\overline{F.T.} = 2.002 - \frac{0.4344 \int \overline{P_f dt}}{1000} + 0.4259 \quad \overline{t(p \geq 250)}$$

The Standard Deviation is defined as:

$$S_{F.T.} = \frac{1}{1000} \sqrt{0.1887 V_s + 186{,}800 V_t}$$

Where V_s is the variance of the integrals and V_τ is the variance of the burning times, at a given temperature.

6.6.2.1 $\int P_f dt$. The integral under the entire pressure time curve measured at the front end of the motor (kilograms sec. per cm^2) with the pressure expressed in kPa.

6.6.2.2 t(P>250). Burning time in seconds, to the nearest millisecond, during which the front motor pressure, as shown on the pressure time curve, exceeds 1723 kPa. It is measured as the interval between the time the pressure first exceeds 1723 kPa and the time when it finally falls below 1723 kPa near the end of burning.

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6.6.3 Early maximum pressure. For the purpose of assessment of pressure time curves, early maximum pressure is the highest pressure during the first 0.20 seconds of burning, or if the trace oscillates, it is the highest pressure reached by the line connecting the mid-points of the initial pressure oscillation.

6.6.4 Late maximum pressure. For the purpose of assessment of pressure time curves, late maximum pressure is the highest pressure after 85 percent of the time interval at the 6.89 Pa level, as measured from the time the initial pressure first exceeds 6.89 Pa until the pressure falls below 6.89 Pa until the pressure falls below 6.89 Pa near the end of burning.

6.6.5 Maximum pressures. For the purpose of assessment of pressure time curves, the maximum pressures are the highest pressures, recorded to the nearest kPa, as scaled from the pressure time curves.

6.6.6 Loss of test results. Malfunctions of components other than the grain in the static fire motor (for example, failure of an igniter to fire) could prevent data compilation for a given round. The minimum sample sizes shown are acceptable in such instances.

6.6.7 Erratic pressure pips (74°C only). Pressure fluctuations or evidence of unstable burning, or both, occurring between 0.20 seconds of initial pressure rise and the 85 percent point as defined in 6.6.4.

6.6.8 Weld line. A structural imperfection detectable under polarized light as an opaque heavy or sharp mark or pattern associated with a high concentration of strain patterns, for example, dark or colored parallel band patterns.

6.6.9 Fissure. For the purpose of this purchase description, a fissure is defined as a cleavage or material separation, generally resulting from a scratch, cut, or similar break in the surface, with measurable width, length, and depth.

6.7 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.8 Subject term (keyword) listing.

Charge, explosive
Inhibitor
Rocket fuel
Rocket motor
Solid fuel

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-18811B (MI)	

3. DOCUMENT TITLE
PROPELLANT GRAIN, MK43, MOD 1

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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